hetmacro Codebook

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January 2026

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CHAPTER 1

Overview

This codebook documents the `hetmacro` package for numerical tools in macroeconomics with heterogeneity. The package is designed to be self-contained and modular, with a foundations-first architecture.

CHAPTER 2

Foundational Modules

2.1 grids.py

- `GridSpec`: specification for one grid dimension
- `make_grid_1d`: flexible spacing and concentration
- `make_grid_nd`: build multi-dimensional grids
- `cartesian_product`, `gridmake`
- `make_asset_grid`, `double_exponential_grid`, `log_spaced_grid`

2.2 quadrature.py

Gaussian and Newton–Cotes quadrature rules.

- `qnwlege`, `qnwcheb`, `qnwnorm`
- `qnwlogn`, `qnwunif`, `qnwbeta`, `qnwgamma`
- `qnwsimp`, `qnwtrap`
- `qnwnorm_mv`

2.3 interpolation.py

- `interp_linear`, `interp_bilinear`
- `get_lottery`
- Spline and Chebyshev tools: `spline_coef`, `cheb_coef`, `cheb_eval`

2.4 optimize.py

Root-finding and optimization wrappers.

- `bisect`, `brentq`, `newton`, `secant`, `broyden`
- `golden_search`, `brent_min`, `nelder_mead`

2.5 markov.py

- `rouwenhorst`, `tauchen`
- `stationary_distribution`, `simulate_markov`

2.6 utils.py

- `crra_utility`, `crra_marginal`, `crra_inverse_marginal`
- `ces_aggregator`
- `compute_fixed_point`, `numerical_derivative`, `numerical_jacobian`
- `tic`, `toc`

CHAPTER 3

Heterogeneous Agent Tools

3.1 `backward.py` and `dp_tools.py`

- `backward_egm`, `backward_vfi`
- `policy_iteration`, `solve_steady_policy`

3.2 `forward.py` and `distributions.py`

- `forward_iteration`, `stationary_distribution`
- `stationary_distribution_ha`, `lottery_indices`

3.3 `steady_state.py`

- `household_steady_state`
- `market_clearing`

3.4 `ssj.py`

- `jacobian`, `step1_backward`, `J_from_F`

CHAPTER 4

Example Model

4.1 `models/aiyagari.py`

- `solve_steady_state`: steady-state solution of Aiyagari model